Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "kinetic energy" and "work" to explain your answer. Sentence starters are optional.

kinetic energy: energy associated with motion

work: energy transferred by a force along a displacement

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

At fixed momentum magnitude, mass becomes **3** times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A **2 kg** cart at **2 m/s** has K=**4 J**. Adding **5 J** of net work gives final K=**9 J**, so final speed is **3 m/s**.

Your turn: Draw before-and-after energy bars and **label the transfer** between them.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

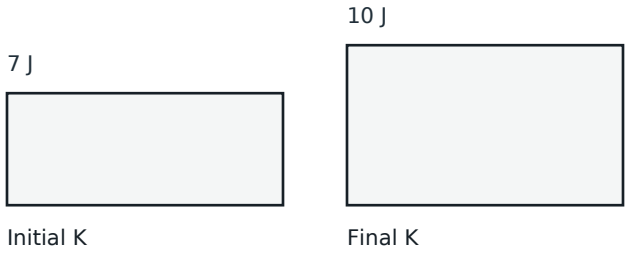
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A lab cart's kinetic energy changes from **7 J** to **10 J**, as shown.
1. **Find the net work**.
2. **Explain how its speed changes** if mass stays constant.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A **4 kg** cart moves at **2 m/s**. The net work on it is **+24 J**. **Complete the energy account**, then **find its final kinetic energy** and speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two equal **4 kg** carts collide elastically along one line. A starts at **+6 m/s**; B starts at rest. **Find both final velocities** in this ideal case.
Use before and after snapshots to organize the momentum account.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** object slows from **6** to **2 m/s**.
1. **Find net work**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **3 kg** object starts at **4 m/s** and receives **-18 J** of net work.
1. **Find final speed**.
2. What is the most negative net work possible before it first stops?
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3